

Coral $\delta^{13}\text{C}$ as Tectonics Proxy & Decadal Interseismic Signatures Qualifying Proposal

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Abstract

Monitoring vertical tectonic displacement over decadal timescales is one of the central challenges in subduction zone geodesy. Conventional geodetic tools; GNSS networks, InSAR, and tide gauges are limited either by sparse spatial coverage, short temporal baselines, or geometric insensitivity in offshore settings. This dissertation proposes and validates a complementary approach: using $\delta^{13}\text{C}$ of coral microatolls as a continuous, high-resolution proxy for relative water depth and thereby for vertical tectonic displacement. The approach is developed and tested using the *Porites* microatoll 12-SPL-A from Vonovono Island, Western Solomon Islands, a region with a well-documented history of large megathrust earthquakes and interseismic deformation.

Preliminary results demonstrate that the low-pass filtered decadal trends in $\delta^{13}\text{C}$ align closely with the coral's morphological record of relative sea-level change, and that a direct $\delta^{13}\text{C}$ -relative water depth calibration is recoverable from samples drilled at known depths. The coseismic uplift of 63 cm associated with the April 2007 M_w 8.1 Solomon Islands earthquake produces a simultaneous $\sim 2\%$ positive excursion in $\delta^{13}\text{C}$, providing a robust calibration anchor. Annual and ENSO-band signals, isolated through Lomb–Scargle spectral analysis and Savitzky–Golay filtering, are clearly separable from the tectonic component.

The dissertation is organized into three research packages. The first focuses on establishing the robustness of $\delta^{13}\text{C}$ as a relative water depth proxy through analysis of the 574-sample 12-SPL-A time series, optimization of a non-linear transfer function, and Suess effect correction. The second extends the proxy to adjacent coral records, including the twin coral 12-SP-A, two corals from North Gizo (12-MBU-A & B), and a multi-core set from Nakaza Island (NAK-A to NAK-H), and investigates analogous $\delta^{13}\text{C}$ signatures in published datasets from tectonically active regions, including the Gulf of Aqaba. The third package integrates the coral paleogeodetic record with modern geodetic observations, combining available GPS campaign data from the New Georgia Group with forthcoming NISAR satellite InSAR data to characterize the spatial distribution of interseismic deformation and evaluate its consistency with the multi-decadal coral record.

Together, these three packages aim to demonstrate that skeletal $\delta^{13}\text{C}$ in microatoll corals can extend the geodetic record of vertical tectonic motion back decades to centuries, providing a new observational window into the interseismic and aseismic phases of the earthquake cycle in the Western Solomon Islands and beyond.

1 Introduction

1.1 Motivation

Understanding the earthquake cycle in its entirety is fundamental to decoding the physics of faulting and improving seismic hazard assessments (Avouac, 2015, Wang, 2024). Within this cycle, the coseismic phase is the more extensively documented component. High-resolution geodetic and seismological data allow for the construction of sophisticated finite-fault models that characterize slip distribution and rupture complexity with high accuracy (Mai and Beroza, 2002, Liu et al., 2024). These models provide a physics-based perspective on how energy is partitioned between radiation and fracture during a rupture (Kanamori and Brodsky, 2004). By characterizing the initial stress state and the distribution of rock properties, the evolution of a rupture can be treated as a deterministic process constrained by the governing laws of friction and elastodynamics.

In contrast, interseismic and aseismic processes remain significantly more enigmatic. While it is now recognized that faults exhibit a spectrum of slip behaviors, ranging from steady creep to episodic Slow Slip Events (SSEs) and “silent earthquakes”, recent findings have challenged our understanding of the preparatory phase of great ruptures (Peng and Gomberg, 2010). Specifically, retrospective analyses of the 2011 Tohoku-Oki (M_w 9.0) and 2014 Iquique (M_w 8.1) earthquakes revealed decadal-scale deformation transients and gradual accelerations of aseismic slip that preceded the mainshocks by years (Mavrommatis et al., 2014, Ruiz et al., 2014). These observations suggest that megathrust earthquakes may undergo a protracted nucleation phase, offering a potential window into the timing of catastrophic failure.

However, capturing these sub-seismogenic changes requires precise, long-term monitoring that is currently hindered by two primary constraints: geodetic resolution and temporal coverage (Bürgmann, 2018, Schwartz and Rokosky, 2007). In the vast majority of global subduction zones, the trench axis lies more than 100 km offshore on average from where the most critical locking occurs (Heuret et al., 2011). Consequently, land-based GNSS networks are geometrically limited in their ability to resolve shallow interseismic locking or differentiate subtle pre-seismic transients from noise due to signal attenuation (Bürgmann and Chadwell, 2014). Researchers are effectively tasked with monitoring millimetre-scale deformation from hundreds of kilometers away. While seafloor geodesy (GNSS-A) offers a technological solution, it remains prohibitively expensive for global-scale, continuous monitoring.

Furthermore, the temporal depth of our geodetic records is critically sparse. In many remote regions, geodetic infrastructure was only recently established, resulting in time series that represent a mere snapshot of a centuries-long cycle. Spaceborne missions such as InSAR provide broader spatial coverage but are similarly constrained by mission duration and environmental noise, including dense vegetation and atmospheric delays, which mask the subtle signals of interseismic loading. To validate the hypothesis that decadal transients are a common precursor to megathrust rupture, these resolution limits must be overcome through strategic geographic selection in areas where fault geometry and observational conditions are optimized.

1.2 Background

1.2.1 Coral Morphology as Paleogeodesy

Measuring coral microatoll growth is a key method addressing many of the limitations described above. This approach provides long, continuous records of vertical tectonic motion with centimeter-scale precision and annual temporal resolution. While their role as sea-level recorders was first recognized by Scoffin and Stoddart (1978), microatolls have since been employed in climate research to reconstruct past ENSO variability (Tudhope et al., 2001, Cobb et al., 2003), in oceanographic studies to track Holocene sea-level trends (Woodroffe and McLean, 1990, Smithers and Woodroffe, 2000), and for tectonic studies, where coral morphology was first applied by Taylor et al. (1987). The field has since extended as a source of geodetic records documenting deformation patterns across multiple earthquake cycles (Philibosian, 2025, & references therein), especially in Sunda Arc (Sieh et al., 2008, Meltzner et al., 2010).

However, coral paleogeodesy based primarily on growth morphology has inherent limitations. In the absence of discrete die-down events recording tectonic uplifts, quantifying exact rates of vertical motion can be challenging (Meltzner et al., 2010). This is particularly evident during periods of subsidence: because coral vertical growth is not strictly depth-dependent as long as the colony remains within the photic zone, the upward growth rate may not accurately reflect the pace of submergence (Woodroffe and McLean, 1990). Consequently, while the method is effective for identifying broad tectonic trends and the direction of motion, the absence of a clear upper-growth constraint during subsidence can lead to underestimation of total displacement.

1.2.2 Stable Isotopes on Corals

Stable isotope ratios measured in coral skeletons are well-established proxies for climate and oceanographic reconstruction, with several decades of application in both fields. The most widely used proxies include $\delta^{18}\text{O}$ for sea-surface temperature (SST) and sea-surface salinity (SSS), $\delta^{13}\text{C}$ for photosynthetic activity and biological process, Sr/Ca for SST. Although these proxies share common environmental drivers, they are not always coupled, and their divergence can itself carry interpretable information. $\delta^{13}\text{C}$ variations in coral skeletons depend on several parameters including SST, SSS, and biological factors intrinsic to

the coral. Among these, light intensity has been identified as one of the primary controls. Goreau (1977) and later on Swart (1983) attributed this pattern, the photosynthetic carbon fixation by observing the photosynthesis rate of the symbiont simultaneously with the coral’s respiration. During photosynthesis, the zooxanthellae symbiont preferentially assimilate the lighter isotope, ^{12}C , from the coral’s internal CO_2 pool. This selective uptake depletes the residual carbon reservoir in ^{12}C and enriches it in ^{13}C , meaning an increased $\delta^{13}\text{C}$. Conversely, Erez (1978) proposed that $\delta^{13}\text{C}$ would decrease with higher light because corals might incorporate depleted metabolic carbon during periods of intense photosynthetic activity.

As the light intensity as well as photosynthetic productivity decreases exponentially with increasing water depth (Dunne and Brown, 1996, Campbell et al., 2007, Obrador and Pretus, 2008), Weber et al. (1976) observed in Caribbean *Montastrea annularis* that skeletal $\delta^{13}\text{C}$ values become more negative with increasing depth for different corals over the uppermost 20 m of the water column, following a non-linear relationship. Weber et al. (1976) also points out that the $\delta^{13}\text{C}$ signal is highly dependent on which morphological part of the coral is being sampled, as different areas have different photosynthetic capacities and it is not clear how relative water depth plays a role on that. One should also consider the effect of seafloor reflection at shallow depths, since light intensity as a function of depth depends on the substrate depth below. The light intensity and water depth function also depends on the depth of the sea floor.

The controlled laboratory experiments conducted by Juillet-Leclerc and Reynaud (2010) on living *Acropora* corals show that the effect of light on skeletal $\delta^{13}\text{C}$ is complex and non-uniform. Rather than displaying a single, consistent response to increased light, individual coral colonies exhibit a dualistic behavior that depends on their specific metabolic and growth responses to light intensity. **$\delta^{13}\text{C}$ Enrichment:** $\delta^{13}\text{C}$ enriched samples are characterized by “low surface extension and high productivity”. In these instances, the coral prioritizes increasing the skeleton density rather than extending its surface. This happens mostly on the lateral sides of the coral, resulting with structures of fiber bundles as well as more positive $\delta^{13}\text{C}$ values. **$\delta^{13}\text{C}$ Depletion:** These samples are associated with the “higher surface or length extension (axial growth) and limited productivity”. This especially occurs on tips of the branches of *acropora* and results with centers of calcification (COC) microstructure and decreased $\delta^{13}\text{C}$.

This duality helps resolve previous debates in the field where contradictory models were proposed by (Goreau, 1977, Erez, 1978, Swart, 1983) as well as complex results delivered by Weber et al. (1976). The experimental results of Juillet-Leclerc and Reynaud (2010) reveal that not only light- $\delta^{13}\text{C}$ but the relationship between $\delta^{13}\text{C}$, $\delta^{18}\text{O}$, light availability, growth rates and skeleton thickening are much more complicated than predicted. The skeletal $\delta^{13}\text{C}$ value is heavily influenced by the isotopic signature of the carbon source used during calcification. Corals may utilize either dissolved inorganic carbon (DIC) from seawater as Goreau (1977) and Swart (1983) suggested or carbon derived from their own metabolic processes, like the respired CO_2 as Erez (1978) was claiming. Because metabolic carbon is typically more depleted in ^{13}C than seawater DIC, the relative contribution of these two sources changes the final isotopic signature of the skeleton.

There is no direct, universal correlation between $\delta^{13}\text{C}$ and net photosynthesis (P_{net}) or respiration (R) when measured in isolation. However, under high light conditions, a significant correlation between $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ emerges, expressed as $\delta^{13}\text{C} = 1.92 \cdot \delta^{18}\text{O} + 3.41$. Consequently, variations in $\delta^{13}\text{C}$ are effectively tied to the balance between linear extension and skeletal thickening, which dictates the relative incorporation of these different microstructures. This is why $\delta^{13}\text{C}$ signals can be highly variable even within the same colony and why simple correlations with environmental factors often fail to account for the full range of isotopic responses (Juillet-Leclerc and Reynaud, 2010).

Linsley et al. (2019) examined the $\delta^{13}\text{C}$ –depth relationship in greater detail using a large, globally distributed dataset of coral samples. Their results corroborate the depth dependence of $\delta^{13}\text{C}$, and although the precise mechanism remains incompletely understood, a linear approximation captures the first-order signal well, yielding a correlation of $r = 0.67$ and a gradient of approximately 1.4‰ per 5 m of water depth.

Connecting the geochemical and tectonic literature naturally raises the question of whether skeletal $\delta^{13}\text{C}$ can serve as a proxy for the relative water depth of the coral colony through time. If so, microatolls could provide continuous, high-temporal-resolution records of vertical tectonic displacement that extend decades to centuries into the past, effectively functioning as natural GPS stations anchored to the seafloor.

2 Hypothesis and Research Questions

The convergence of coral geochemistry and tectonic geodesy outlined above motivates the central hypothesis of this dissertation: that skeletal $\delta^{13}\text{C}$ in corals records relative water depth through time and can therefore serve as a continuous, high-resolution proxy for vertical tectonic displacement. To test this hypothesis and explore its implications, this project will address the following questions:

1. Is $\delta^{13}\text{C}$ a viable proxy for coral based paleogeodesy?
2. If so, what does the $\delta^{13}\text{C}$ record reveal about vertical deformation during interseismic periods in the Western Solomon Islands as a case study?

3 Preliminary Work

3.1 Field Work

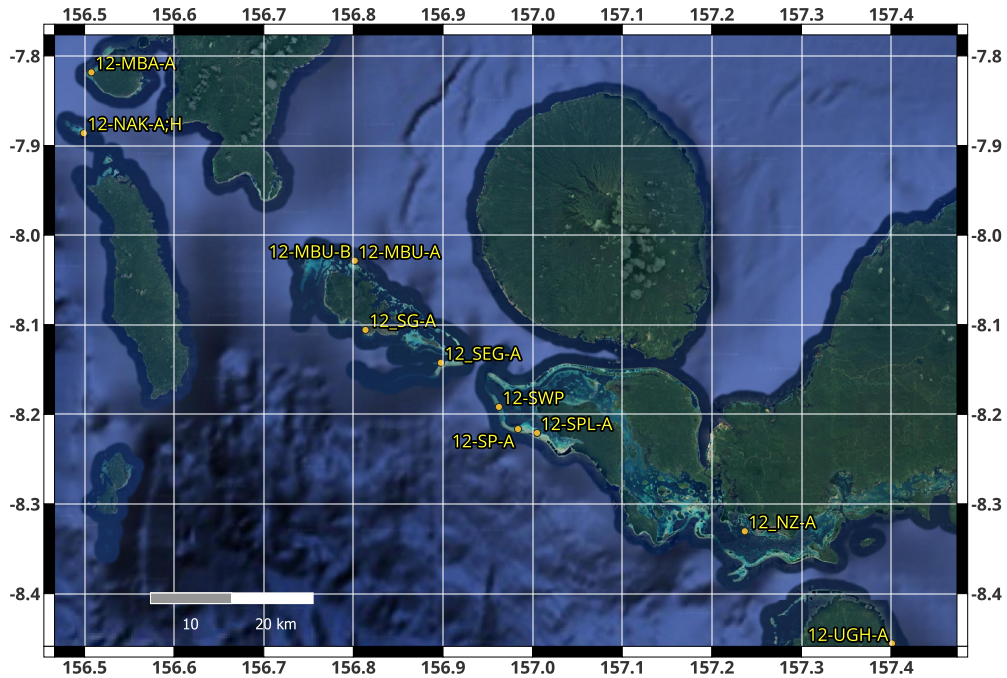


Figure 1: Overview map of New Georgian Islands with coral samples collected during 2012–2013.

Taylor et al. (2008) documented the surface deformation caused by the M_w 8.1 April 1 2007 Solomon Islands earthquake (International Seismological Centre, 2026) across the region. Sixty-five field measurements span from 2.5 m of uplift to 0.7 m of subsidence over an area of approximately 150×70 km. Similarly, Miyagi et al. (2009) reported more than 2 m of line-of-sight (LOS) coseismic deformation from the same event based on ALOS PALSAR observations. The Solomon Islands represent a data-scarce environment characterized by large gaps in observational coverage, an absence of in situ measurements, and a remote, difficult-to-access study site. There are no permanent GPS stations on the New Georgia island chain near the epicentre, and only two operate in the Solomon Islands as a whole. Limited campaign GPS data exist, but their temporal coverage is much shorter and the network is considerably sparser than typical permanent geodetic networks.

To extend the geodetic record, Fred Taylor and his team collected 15 microatolls in 2012–2013 from New Georgian Islands Group (NGG) from Ranongga Island to Vella Lavella Island in the west, then eastward to Ghizo and surrounding smaller islands, and onward to the western tip of Parara Island (Fig.1). Stable isotope analysis was initiated specifically on corals 12-SPL-A and 12-SP-A, recovered from the inner part of the lagoon on the western end of Vonovono Island (referred to in Taylor et al. 2008

and locally known as “Parara”). The coordinates for 12-SPL-A are 8.221° S, 157.005° E and for 12-SP-A are 8.216° S, 156.984° E. Both microatolls are approximately 4 m in diameter with very similar surface topography; nearby microatolls are visually comparable. 12-SPL-A was uplifted at least 63 cm and 12-SP-A 44 cm; 12-SPL-A survived the event while 12-SP-A died completely. Post-seismic morphological recovery in 12-SPL-A shows 7 cm of vertical growth, considering that the alive coral surface lay 10 cm below sea while sampling, there is at least 17 cm of post-seismic subsidence in the years following the earthquake.

3.2 Sampling

Two separate sets of drilling paths were conducted on 12-SPL-A. In both cases, a temporally focused horizontally oriented time series and multiple shorter transects for depth- $\delta^{13}\text{C}$ calibration were collected (Fig. 2). The first drilling paths were conducted in 2023 and its primary results, which show a compelling agreement with InSAR observations from the region, will be published in Kendall et al. (Under Review). 118 continuous samples were collected covering the 2007 M_w 8.1 event, together with 68 calibration samples from seven different depths.

3.3 Age Model

The second and main drilling paths (red & yellow paths in Fig. 2) for the present study took place in late 2024. I collected 574 samples from eight horizontally parallel paths to provide temporal coverage of the tectonic history, along with 140 samples from five depths for the depth- $\delta^{13}\text{C}$ calibration. The paths were oriented horizontally to capture temporal changes in stable isotope ratios driven by vertical tectonic motion. Although this differs from the conventional vertical sampling strategy, sampling in either direction does not alter isotope ratios (e.g. Sr/Ca) provided the growth axis is respected, as demonstrated by DeLong et al. (2013).

I developed a climate-reference age model building on established methods from prior coral chronology studies (Lawman et al., 2020). I prioritized anchoring the age model to local climate data and then extrapolating the chronology beyond the period for which the climate reference is available. X-ray growth-band counting was deliberately excluded as the primary age indicator because density bands on equatorial *porites* can be ambiguous; however, X-ray images were used to establish spatial connections between sampling paths within the coral slab and between the samples and the coral morphology.

Each slab section was X-ray scanned individually after isotope sampling. Indicator notches were carved at the end of every six samples to localize each sample precisely on its path in the X-ray image. Adjacent slab sections were digitally registered to reconstruct the continuous slab. This spatial registration provided the relative stratigraphic position of each sample and allowed paths to be anchored to their neighboring paths.

As the earlier age models confirmed, density bands derived from X-ray radiography can yield inaccurate absolute ages. In our initial growth-band count, common errors associated with false or stress bands produced an age estimate approximately ten years too old for the youngest sample in the core. Although absolute age counting from density bands can be unreliable, their unique barcode-like spatial structure is well suited to temporal-spatial correlation between isotope records and morphological observations. X-ray bands from 12-SPL-A were therefore also used as a reference for 12-SP-A.

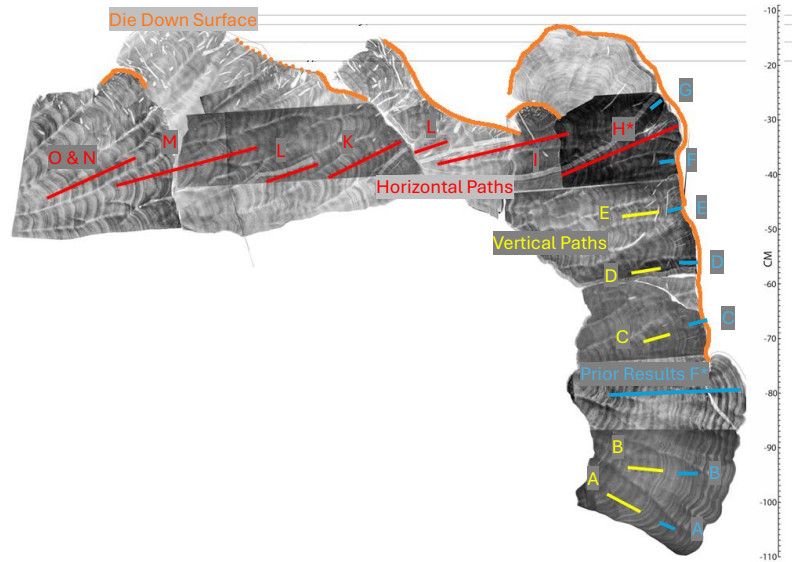


Figure 2: All X-Ray images patched together with all the sampling paths. Blue lines indicate the first drilling paths (Kendall et al. Under Review), the red and yellow lines indicate the second drilling paths. Orange straight lines indicate die-down surfaces and dotted orange line is possibly the erosion surface.

Because we are working with a recent coral, our age model is based on the pseudo-coral forward model developed by Thompson et al. (2011). Tie points were established by matching synthetic and measured $\delta^{18}\text{O}$ signals. The forward model expresses coral $\delta^{18}\text{O}$ as a linear combination of sea-surface temperature (SST) and sea-surface salinity (SSS): $\delta^{18}\text{O}_{\text{synthetic}} = \alpha_1 \text{SST} + \alpha_2 \text{SSS}$, using the geographically calibrated coefficients proposed by Thompson et al. (2011): $\alpha_1 = -0.22 \text{‰}/^\circ\text{C}$ and $\alpha_2 = 0.80 \text{‰}/\text{‰}$. Monthly SST and SSS fields were taken from the ORAS5 ocean reanalysis (Copernicus Climate Change Service, 2021) for the period 1958–2007.

Annual cycles were identified in the coral $\delta^{18}\text{O}$ record and matched one-to-one to the corresponding annual cycles in the synthetic climate reference. This produced a set of cycle-boundary anchor points in the X-ray-based pseudo-year space, each mapped to a calibrated calendar year. These anchor pairs were used to construct a piecewise-linear warp function that converts raw pseudo-years based on sampling order into calibrated calendar years. The resulting age model spans 1958–2007 and yields a Pearson correlation of $r = 0.729$ against the climate reference. For the interval predating the available climate record (before 1958), tie points were assigned based on the long-term annual climatological average of the pseudo-coral signal.

3.4 Suess Effect Correction

Linsley et al. (2019) investigated the role of the Suess effect on coral skeletal $\delta^{13}\text{C}$ parallel to the role of relative water depth. They explicitly acknowledge that the decline in atmospheric $\delta^{13}\text{C}$ values directly impacts surface ocean dissolved inorganic carbon (DIC) $\delta^{13}\text{C}$ through the gas exchange of CO_2 , a process they define as the “oceanic $\delta^{13}\text{C}$ Suess effect”. To account for this, I also subtracted the amount of $\delta^{13}\text{C}$ depletion caused by the Suess effect from each sample by using the same time series as Linsley et al. (2019). This is a methodological simplification rather than a perfect replication of local ocean chemistry.

3.5 Relative Water Depth & $\delta^{13}\text{C}$

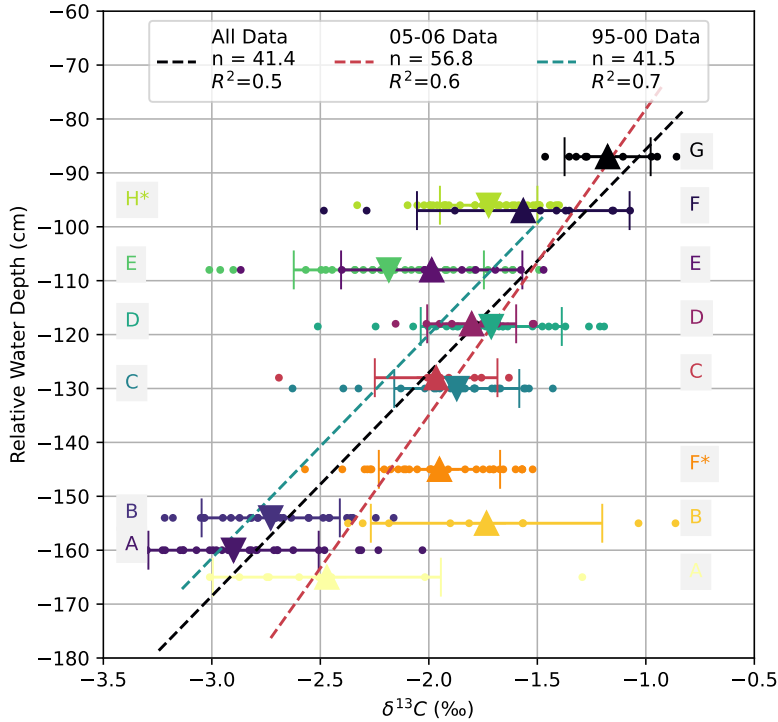


Figure 3: Relationship between relative water depth and skeletal $\delta^{13}\text{C}$. Small dots show measurements; large markers show the path mean $\pm 1\sigma$. Upward-pointing triangles ($\blacktriangle$) denote paths of the ages between 1995–2000 from the second drilling campaign. Downward-pointing triangles ($\blacktriangledown$) denote paths of the ages 2005–2006 from the first drilling campaign, to which a $+0.25\text{‰}$ Suess correction has been applied.

Based on the die-down surface of the 12-SPLA (Fig 2), the M_w 8.1 2007 earthquake (International Seismological Centre, 2026) exposed 63 cm from the top of the coral above the sea level. On the other hand the coseismic change in $\delta^{13}\text{C}$ is $\sim 2 \text{‰}$, yielding an estimated $\delta^{13}\text{C}$ -water depth relationship of $\sim 31.5 \text{ cm}/\text{‰}$ from the morphology.

To investigate the relationship between $\delta^{13}\text{C}$ and relative water depth directly, a dedicated sampling strategy was adopted. To minimize temporal variability, samples must be taken from the same growth band but at different depths. In practice, a short horizontal sampling patch is extracted at each target depth and the ages are subsequently matched across depths. During the first drilling campaign, an average of 10 neighboring samples were collected per depth; this was increased to an average of 28 samples per depth during the second campaign. For the depth- $\delta^{13}\text{C}$ calibration, a correction is necessary because the two datasets being compared were calcified at different times (1995–2000 and 2005–2006). The

younger dataset was therefore corrected by $+0.3\text{‰}$, corresponding to the difference in the Suess effect (Section 3.4) between 1997 and 2006 following Linsley et al. (2019). Path H from the second drilling campaign is combined with the 1995–2000 results; path F from the first drilling campaign is combined with the 2005–2006 results. The linear regression reveals a positive correlation between $\delta^{13}\text{C}$ and relative water depth. However, scatter increases noticeably with depth, which may reflect greater seasonal variability in temperature or salinity at depth. A further complication is the possible non-linearity of the relationship; the observed spread across such a small depth range is considerably larger than reported by Linsley et al. (2019), which warrants further investigation.

3.6 Time Domain Analysis

3.6.1 $\delta^{13}\text{C}$ & $\delta^{18}\text{O}$ Trends

Both $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ capture a range of processes simultaneously: annual and ENSO-driven climatic cycles alongside decadal-scale signals related to vertical tectonic motion. The two isotope ratios also differ in their sensitivity to the same processes, which makes their joint interpretation necessary.

The $\delta^{13}\text{C}$ record is divided into eight roughly decadal segments, each described by a linear trend (Fig. 4). The earliest transition occurs around 1945, where $\delta^{13}\text{C}$ shifts from a near-flat trend to a marked positive inclination that persists until approximately 1951. At the same inflection point, $\delta^{18}\text{O}$ drops abruptly by roughly twice the annual amplitude; nevertheless, the gradual increasing trend in $\delta^{18}\text{O}$ continues through 1953. Following this, $\delta^{18}\text{O}$ enters a remarkably stable, near-zero trend that lasts for 18 years until late 1970. This interval coincides precisely with the steepest increase in $\delta^{13}\text{C}$ and its peak in October 1970. A key observation is that the correlation between detrended $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ remains statistically significant throughout this period despite their strongly diverging long-term trends. This supports the in-

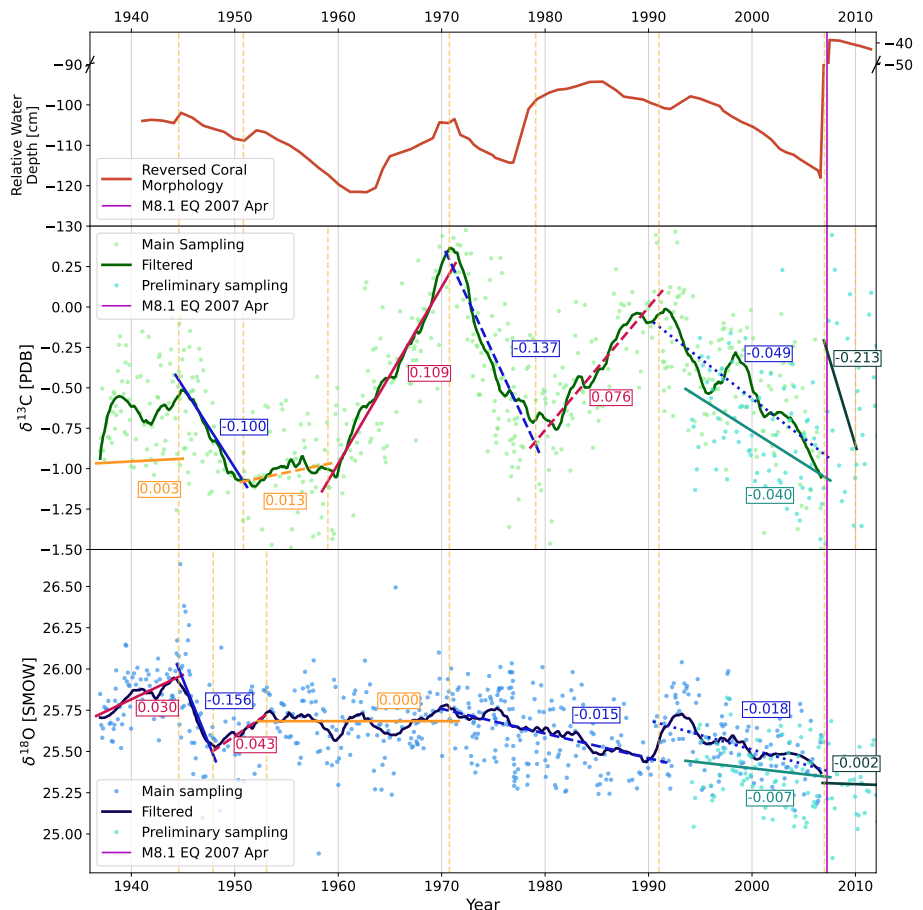


Figure 4: Piecewise linear trend analysis of $\delta^{13}\text{C}$ (middle panel, VPDB) and $\delta^{18}\text{O}$ (lower panel, VSMOW) for coral 12-SPL-A, together with the reversed coral morphology expressed as relative water depth (upper panel). Green and blue dots show the main sampling record (~ 1937 –2007); cyan dots show the preliminary dataset spanning 1997–2011. Coloured line segments are piecewise linear fits to distinct trend intervals, with slopes in ‰ yr^{-1} labeled in matching colored boxes. The dark overlaid line, labeled “filtered” in each isotope panel, is the Savitzky–Golay long-term trend (window = 73 months, polynomial order 3), used to isolate decadal-scale signals from higher-frequency variability. Vertical dashed orange lines mark the segment boundaries shared across all panels, highlighting the close agreement between the $\delta^{13}\text{C}$ trends and the relative water depth record. Because the coseismic uplift occupies a large portion of the morphology y -axis, the axis is broken and the interval between -90 and -50 cm is omitted. The magenta vertical line indicates the April 2007 M_w 8.1 Solomon Islands earthquake.

terpretation that the decadal-scale trends in $\delta^{13}\text{C}$ have a non-climatic, environmental origin, which is explored in detail in later sections. After a steep decline through 1979, $\delta^{13}\text{C}$ begins a new increasing trend that continues until 1991. Concurrently, a sustained decreasing trend in $\delta^{18}\text{O}$ spanning approximately 20 years terminates at the same year, which coincides with the onset of the 1991–92 El Niño. From 1991 onward, both signals show a declining trend leading up to the 2007 earthquake.

Despite considerable high-frequency variability in the monthly data, a three-point moving average reveals that the annual cycle is the dominant feature in both $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$. For $\delta^{18}\text{O}$, a clear seasonal peak is observed between May and November. The $\delta^{13}\text{C}$ seasonal pattern is less regular: the signal typically reaches its minimum in June and its main peak in October. However, there are intervals where the annual signal weakens or becomes indistinguishable against the background of strong decadal trends, particularly in $\delta^{13}\text{C}$. $\delta^{18}\text{O}$ is a well-established indicator of SST and SSS variability, and the measured $\delta^{18}\text{O}$ record aligns well with both. A Lomb–Scargle periodogram of $\delta^{18}\text{O}$ shows that the annual signal is suppressed only during the 1992 ENSO event and weakens around the 1945 drop. $\delta^{13}\text{C}$ behaves less consistently, reflecting its greater sensitivity to shorter-term environmental changes relative to $\delta^{18}\text{O}$. The annual signal in $\delta^{13}\text{C}$ is present through most of the record, but is interrupted during several year-long intervals that coincide with strong long-term trends, notably the period following the 1945 drop and the region around the 1970 peak.

3.6.2 Morphology

The most striking feature of the $\delta^{13}\text{C}$ signal in the time domain is its strong alignment with the coral’s macroscopic morphology (Fig. 4). This result was expected due to experimental observations of Juillet-Leclerc and Reynaud (2010) on the effect of growth rates on skeletal $\delta^{13}\text{C}$. Both abrupt die-down events and sustained subsidence or uplift episodes visible in the coral morphology are reflected in $\delta^{13}\text{C}$ levels, and some cases in $\delta^{18}\text{O}$ as well. The primary motivation for this study derives from observations made following the M_w 8.1 2007 earthquake. Results from the first drilling campaign (Kendall et al. Under Review) demonstrate clearly that this event produced significant changes in both isotope ratios along transect H’: $\delta^{13}\text{C}$ rose sharply after the event and then declined characteristically over the following months, while $\delta^{18}\text{O}$ showed no marked increase but its annual periodicity disappeared abruptly. The 72 cm coseismic uplift corresponds to approximately 2 ‰ and at the end of the postseismic response on 2010 $\delta^{13}\text{C}$ reaches nearly down to initial values before the event.

Another clear die-down surface occurs at approximately 1945 in both 12-SPL-A and 12-SP-A, accompanied by an unexpected negative excursion in $\delta^{18}\text{O}$. Notably, the periodicity and long-term trend in $\delta^{18}\text{O}$ remain unchanged after this event. A declining trend in $\delta^{13}\text{C}$ of approximately five years duration follows the 1945 event and is captured independently by two different drilling paths (end of path N and start of path M). The independent samples from both paths are mutually consistent and align with the morphological records of both corals. The decline persists until approximately 1950, during which time the annual periodicity is substantially disrupted before a temporary recovery around June 1952. Between 1950 and 1959 a decadal-scale stable trend is observed while the morphology continues to grow, indicating either a relative sea-level rise or a sustained period at greater relative water depth. A marked increase in $\delta^{13}\text{C}$ begins around 1959, approximately four years before the morphological trend reverses its downward direction. Both the morphology and $\delta^{13}\text{C}$ reach their maximum at the beginning of 1970 and subsequently decline. The apparent change in morphological character in 1976 is tentatively attributed to post-sampling surface erosion, as growth bands appear perpendicular to the slab surface in the upper section. The only interval in which morphology and $\delta^{13}\text{C}$ diverge is 1985–1993, a period that encompasses two major ENSO events and during which $\delta^{13}\text{C}$ shows a marked negative correlation with the Honiara tide gauge record and with SST (Fig. 5). This is followed by approximately 18 months of stability and then strong subsidence leading up to the 2007 event. During this subsidence interval, vertical growth rates substantially exceeded horizontal growth rates, and the resulting asymmetry is visible in Fig. 2. The reduced horizontal extension during this period complicated both annual growth-band counting and the matching of $\delta^{18}\text{O}$ to the climate reference. This difference between vertical and horizontal growth rates reflects a change in growth mode, as discussed in Section 1.2.2, and the X-ray patterns observed here are consistent with the experimental observations of Juillet-Leclerc and Reynaud (2010).

3.6.3 External Data Sets

Three external datasets were used as climate references throughout the time-domain, spectral, and rolling-correlation analyses. Monthly sea-surface temperature (SST) and sea-surface salinity (SSS) fields were obtained from the ORAS5 global ocean reanalysis (Copernicus Climate Change Service, 2021). ORAS5 provides monthly gridded fields at 0.25° horizontal resolution from January 1958 to the present, making it the longest available observational-quality ocean reanalysis and therefore well suited for calibration against the full length of the coral record. SST and SSS were extracted from the grid cell nearest to the study site (8.22° S, 157.00° E) and used to construct the synthetic $\delta^{18}\text{O}$ reference signal following the pseudo-coral forward model of (Thompson et al., 2011).

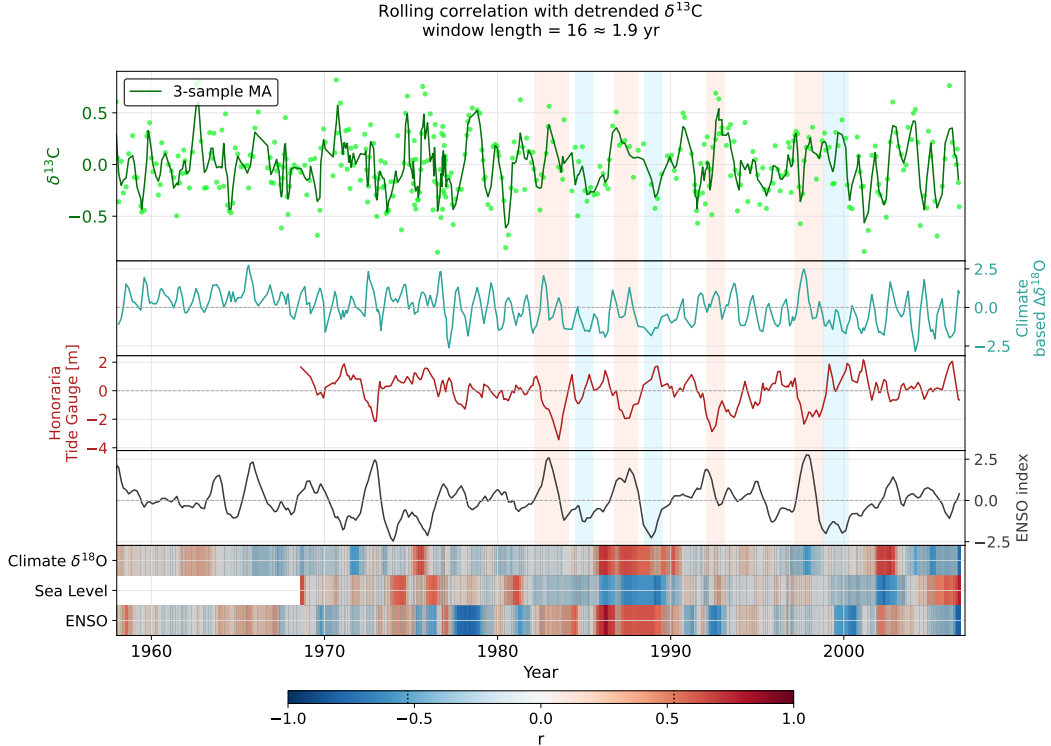


Figure 5: Rolling correlation between Savitzky-Golay detrended skeletal $\delta^{13}\text{C}$ and three climate drivers over the coral record. **Top panel:** raw $\delta^{13}\text{C}_{\text{no-tec}}$ (dots) with 3-sample centered moving average (solid line). **Middle panels:** climate-based $\Delta\delta^{18}\text{O}$ derived from ORAS5 reanalysis sea-surface temperature and salinity (Thompson et al., 2011); Honiara tide gauge sea level anomaly (m); and the ENSO index (all z-scored). **Bottom panel:** Pearson rolling correlation coefficient r between detrended $\delta^{13}\text{C}$ and each driver, computed with a centered window of 16 samples (≈ 1.9 yr).

Sea-level variability at the study site was characterized using monthly tide gauge records from the Honiara station (Guadalcanal, Solomon Islands), obtained from the Joint Archive for Sea Level (Caldwell et al., 2001) maintained by the NOAA National Centers for Environmental Information. Honiara is the nearest long-running tide gauge to the New Georgia Group, situated approximately 200 km to the east of the study site. The record commences in 1968, which accounts for the data gap in the sea-level driver panel prior to that year (Figs. 5). Monthly anomalies were computed by subtracting the long-term monthly mean climatology from the raw observations.

ENSO variability was represented by the Oceanic Niño Index (ONI), a three-month running mean of SST anomalies in the Niño 3.4 region (5° N– 5° S, 170° – 120° W) relative to a centered 30-year base period (National Centers for Environmental Information (NCEI), 2026). El Niño and La Niña episodes were defined as periods during which the ONI exceeded $\pm 0.5^\circ\text{C}$ for at least five consecutive overlapping three-month seasons, following the standard NOAA classification. The resulting event windows are indicated by orange and blue shading, respectively, in all rolling-correlation and time-series figures.

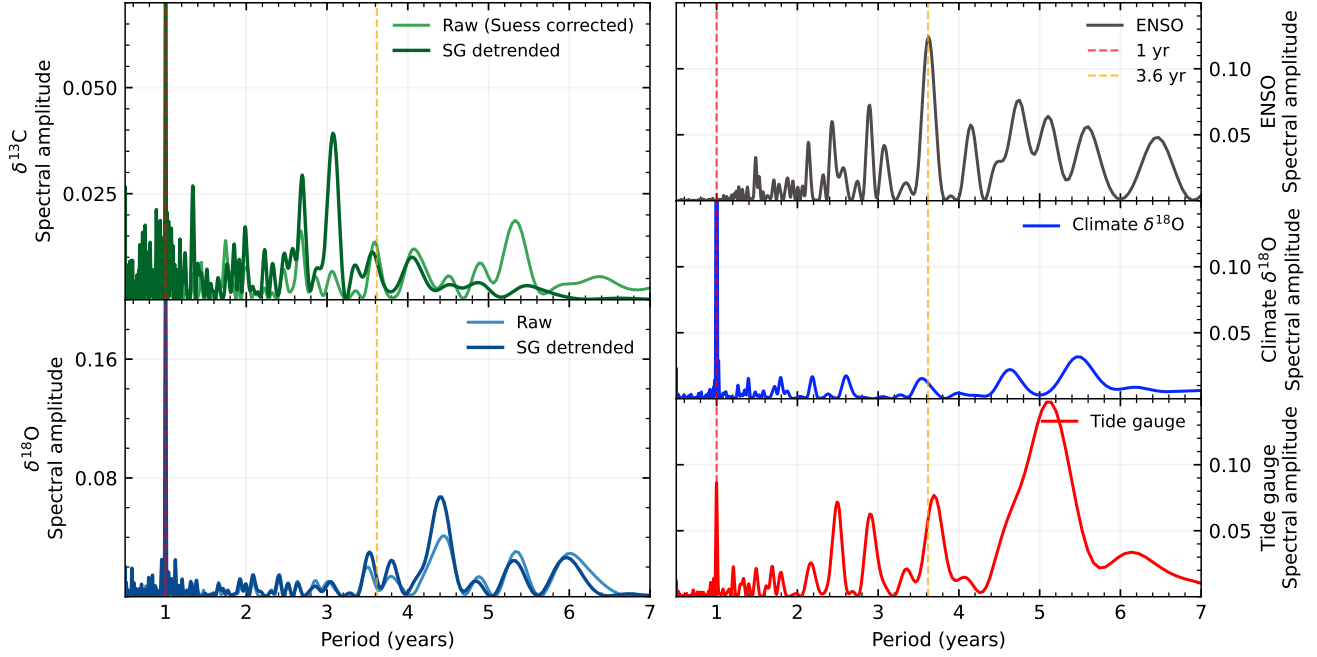


Figure 6: Lomb–Scargle power spectra of coral skeletal isotope records (1958–2007, left column) and three climate drivers (right column) for the SAG microatoll record. Left panels show raw and Savitzky–Golay detrended $\delta^{13}\text{C}$ (top) and $\delta^{18}\text{O}$ (bottom). Right panels show spectra of ENSO index, the ORAS5-derived synthetic $\delta^{18}\text{O}$ and Honiara tide gauge sea level. Dashed vertical lines mark the 1-year (red) and 3.6-year (orange) periods.

3.7 Spectral Domain

As is common for palaeoclimatic records, sampling intervals in this study’s dataset are not perfectly uniform. I therefore employed two complementary spectral approaches: (1) resampling the data to a regular 2-month grid prior to fast-Fourier transform (FFT) analysis, and (2) Lomb–Scargle periodogram analysis (Press and Rybicki, 1989) applied directly to the raw, unevenly sampled time series.

Both approaches consistently reveal three groups of dominant periodicities in both isotope signals: (i) Annual. (ii) ENSO band. (iii) Low-frequency components.

3.7.1 Annual Climatic Signal

The annual signal reflects the yearly variability driven primarily by the solar cycle, which modulates SST directly and influences $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ indirectly through changes in cloudiness and precipitation (Fig. 6). Within the equatorial belt, seasonal SST variations are of order $\pm 3^\circ\text{C}$, which alone is insufficient to account for the full annual amplitude in either isotope ratio; the rainy season and associated cloudiness exert an additional influence on both signals. Monthly averages of $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ exhibit a 1–2 month offset in their respective troughs, and their seasonal waveforms differ in character, indicating that they are governed by partially distinct processes. $\delta^{18}\text{O}$ is markedly more robust and shows a clear maximum between May and December, with a relatively stable low-amplitude interval from January to April. $\delta^{13}\text{C}$, by contrast, is more dynamic and noisier, with a major peak in September and a secondary peak in March. This seasonal behaviour of $\delta^{13}\text{C}$ is broadly consistent with the monthly SST climatology. $\delta^{18}\text{O}$ is negatively correlated with SST, whereas $\delta^{13}\text{C}$ shows a slight positive relationship.

3.7.2 ENSO Signal

Four major ENSO events in the observational record are also captured by the Honiara tide gauge: 1982/02–1984/01, 1986/09–1988/01, 1992/01–1993/01, and 1997/02–1998/10. All four events are expressed as significant positive excursions in $\delta^{18}\text{O}$. The response of $\delta^{13}\text{C}$ is more complex. La Niña events (positive ENSO index) are associated with increased SST, slightly elevated rainfall, and lower sea levels in the western Pacific (Murphy et al., 2014). El Niño events produce more muted effects in the region; moreover,

El Niño events themselves fall into two subcategories—Warm Pool events, characterised by slightly positive SST and slightly negative rainfall anomalies, and Cold Tongue events, with subtly negative SST and stronger negative rainfall anomalies. Because these parameters exert opposing influences on $\delta^{13}\text{C}$, ENSO is not always unambiguously identifiable in the $\delta^{13}\text{C}$ time series. Nevertheless, events in 1977, 1987, and 2005 are identifiable through their correlation with SST, sea level, and the ENSO index. The Lomb–Scargle periodogram reveals several statistically significant peaks consistent with ENSO forcing: a dominant period at 3.1 years and secondary peaks at 2.6 and 4.0 years (Fig. 6).

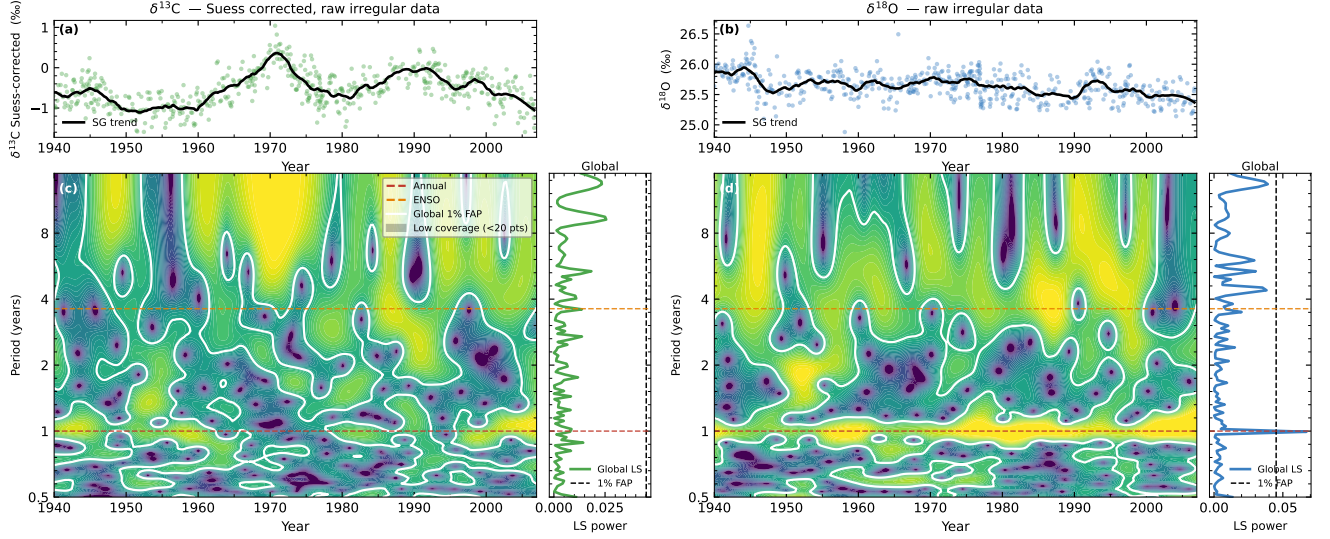


Figure 7: Sliding-window Lomb–Scargle (SWLS) spectrograms of Suess-corrected $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ from the 12-SPL-A record. **(a–b)** Raw isotope time series with the low pass trend overlaid. The full signal, including the low-frequency tectonic component, is retained as input. **(c–d)** SWLS spectrograms. At each 3-month time step a Gaussian-weighted Lomb–Scargle periodogram is computed directly on the original irregular samples within a window of standard deviation $\sigma = 2\text{ yr}$, without resampling. Measurement weights decay exponentially and are passed to the estimator as inverse uncertainties, so that samples near the window centre constrain the fit more strongly. Colour intensity represents $\log_2$ -normalised LS power. The annual cycle (1 yr, red dashed line) is continuously present throughout the record in $\delta^{18}\text{O}$ and is intermittently visible in $\delta^{13}\text{C}$.

4 Discussion and Conclusion

$\delta^{13}\text{C}$ is a complex climate proxy and, as a result, has not been widely adopted in palaeoclimate research. However, in the context of coral paleogeodesy it may be the only proxy available. Consequently, the proxy holds great potential to unlock new analytical opportunities for paleogeodesy. The biological and marine geochemical processes that complicate $\delta^{13}\text{C}$ at monthly to annual timescales appear to average out at decadal periods, leaving the long-term signal dominated by a different, more stable forcing. The main limitation of the proxy, within the scope of the present preliminary results, concerns the $\delta^{13}\text{C}$ –depth relationship in very shallow water and its behavior at the decimeter scale. It remains unclear how the $\delta^{13}\text{C}$ –relative depth conversion operates at depths shallower than approximately 50 cm or deeper than roughly 3 m. Current understanding suggests that the relationship is non-linear and that the sign of the correlation may reverse in the shallowest part of the water column. Having isolated the annual and ENSO components of the $\delta^{13}\text{C}$ signal, and guided by the observed $\delta^{13}\text{C}$ –depth relationship, I interpret the low-pass filtered decadal trends in $\delta^{13}\text{C}$ as tectonic relative depth variations. No other known forcing produces such coherent, long-term trends in coral skeletal $\delta^{13}\text{C}$. The proximity of the study site to the subduction trench further supports this interpretation, accounting for both the amplitude of the observed rates and the frequency of trend reversals. One plausible mechanism is temporal variation in the degree of coupling between the overriding and subducting plates, driving decadal-scale vertical displacement. Under this interpretation, the prolonged uplift intervals recorded in the $\delta^{13}\text{C}$ signal would represent aseismic slow-slip events lasting multiple years, analogous to those documented in the Hikurangi Subduction Zone by Wallace (2020).

4.1 Summary of Evidence Supporting the Proxy

Three independent lines of evidence support the use of $\delta^{13}\text{C}$ as a paleogeodetic proxy in this setting: (i) The 2007 earthquake produced a coseismic uplift that is simultaneously recorded as a sharp positive excursion in $\delta^{13}\text{C}$, providing a direct calibration point between known vertical displacement and isotopic response. (ii) The low-pass filtered $\delta^{13}\text{C}$ signal shows strong agreement with the morphological records of both 12-SPL-A and 12-SP-A, two independent coral colonies with independently documented stratigraphic histories. (iii) The preliminary spectral and time-domain analysis demonstrates that the annual and ENSO climate signals can be effectively separated from the low-frequency tectonic component, leaving a residual that is internally consistent and geophysically interpretable.

5 Research Package 1: Robustness and Validation of the $\delta^{13}\text{C}$ –Relative Water Depth Relationship

Building on the preliminary results presented in Section 3, the following steps remain to complete the robustness analysis: (i) Optimizing a non-linear transfer function for the $\delta^{13}\text{C}$ –relative water depth relationship. (ii) Applying further tests to the Suess-effect-corrected data. (iii) Finalizing the manuscript and proposal

The first research package will introduce $\delta^{13}\text{C}$ as a relative water depth proxy to the paleogeodesy community and establish its robustness by demonstrating its strong correlation with morphology and the separability of the tectonic component from the climate component of the $\delta^{13}\text{C}$ signal.

6 Research Package 2: Further Analysis and Validation Using Adjacent Corals and Literature Records

6.1 Sampling of 12-SP-A

To validate the results obtained from 12-SPL-A, the first focus of the next phase will be the twin coral 12-SP-A. The prepared slabs (1–2 cm thickness) are currently stored at the UTIG warehouse. Based on the sampling and analysis experience gained from 12-SPL-A, several preparatory steps are essential before drilling begins. Based on the age model developed for 12-SPL-A, it is clear that a higher sampling resolution is necessary for improved chronological precision. Therefore, the drilling interval should be reduced from 1.5 mm to 1.0 mm, which increases both laboratory time and analysis costs. A practical compromise is to apply the finer 1.0 mm spacing selectively, for example, across intervals of particular tectonic interest, while retaining 1.5 mm spacing elsewhere. After sampling, the slabs should be X-ray scanned again using the same standardized settings, so that the positions of the drilled paths are precisely digitized for the final age model. The effect of sampling direction on stable isotope ratios in coral slabs remains a debated topic in the palaeoclimate community. To address this, I plan to replicate the experiment of DeLong et al. (2013) by comparing $\delta^{13}\text{C}$ values measured along the same growth rings sampled in both the vertical and horizontal directions. Based on the small offsets observed between overlapping sections of the horizontal paths in the present dataset, I expect minor offsets or subtle trend differences between the two directions, but the overall signal structure should be preserved. This comparison would provide an additional test of the robustness of $\delta^{13}\text{C}$ as a proxy that is independent of sampling orientation. In addition to $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$, Sr/Ca ratios will be measured to improve the accuracy of the age model and to better separate climatic from tectonic signals. Because 12-SP-A was living at a shallower depth than 12-SPL-A, its $\delta^{13}\text{C}$ values are expected to be more positive on average and to show greater variability in response to vertical motion, since light intensity changes more steeply per unit depth in the upper water column. At first order, the long-term $\delta^{13}\text{C}$ trend should resemble that of 12-SPL-A, consistent with the similar morphological histories of the two corals. The $\delta^{18}\text{O}$ records are also expected to agree closely, most likely with a near-constant offset, which should considerably simplify the age modeling of 12-SP-A.

6.2 Adjacent Corals

Fred Taylor and colleagues collected dozens of samples across the SW Pacific during his 40-year career. Thanks to his efforts, early-career scientists have access to a unique collection that includes not only coral

slabs but also multiple unpublished laboratory results, including $\delta^{13}\text{C}$ analyses. Within the scope of this dissertation, two main criteria guide the selection of additional study sites: (1) **Tectonically stable regions:** Passive Pacific hotspot islands and atolls would serve as a control group to test whether $\delta^{13}\text{C}$ records from tectonically quiescent settings show the same $\sim 2\text{‰}$ decadal amplitude changes observed near subduction zones. These sites do not need to be in close proximity to the primary study area, provided that climate data are available to account for the main climatic drivers of $\delta^{13}\text{C}$ variability. (2) **Sites within the same tectonic system:** To validate the tectonic events identified in 12-SPL-A and 12-SP-A, additional sites should lie within close proximity to the primary study area. Both Taylor et al. (2008) and Kuo et al. (2016) document strong tectonic heterogeneity and spatial variability in vertical motions across the NGG; consequently, even nearby corals may show significant divergence if the deformation source is localized.

6.2.1 12-MBU-A & 12-MBU-B

The north-western coast of Ghizo Island lies near the hingeline between coseismic uplift and subsidence during the April 2007 earthquake (Taylor et al., 2008). Two *Porites* microatolls, 12-MBU-A and 12-MBU-B, were sampled from a small barrier island on 8.028°S , 156.802°E approximately 50 m apart in very shallow water; both were living at the time of collection and lay approximately 10 cm below sea level. Neither coral shows any evidence of uplift associated with the 2007 earthquake, consistent with their location near the deformation hinge. However, both record an uplift event peaking around 1970 followed by rapid subsidence until approximately 1981, which aligns well with the morphological record of 12-SP-A. These morphological observations raise the question of what the $\delta^{13}\text{C}$ signal looks like during the period of continuous upward growth observed since the mid-1980s.

6.2.2 NAK-A to NAK-H

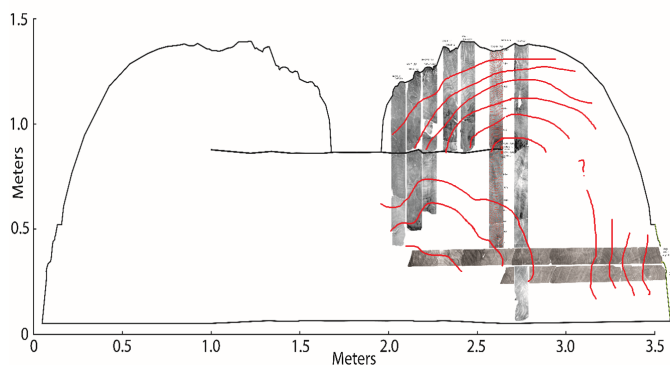


Figure 9: Profile sketch and X-ray images of cores collected from coral 12-NAK, comprising five vertical and two horizontal cores. Red lines indicate the dominant growth band orientations, highlighting the unconformity.

one coincides with the M_w 7.5 earthquake of 17 August 1959, whose epicentre was located approximately 30 km north-west of the northern end of Ranongga Island. That event caused approximately 60 ft of

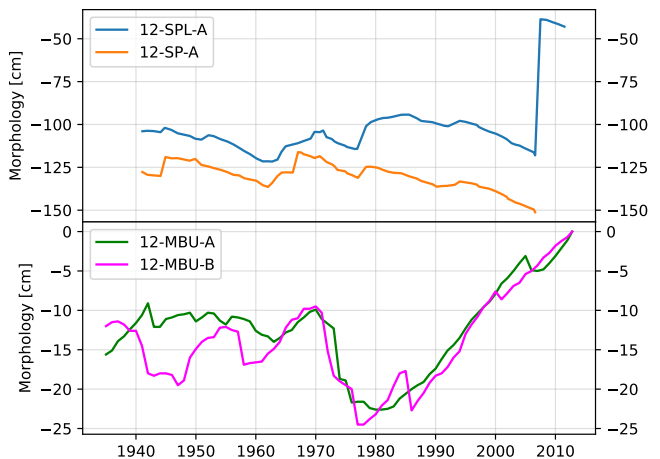


Figure 8: Morphological records of four microatoll corals from the Western Solomon Islands spanning ~ 1935 –2012. Upper panel: Vonovono Island corals. Lower panel: North Ghizo Island corals.

Coral NAK was recovered from Nakaza Island, a 200 m-long island surrounded by an elliptical atoll with a major axis exceeding 3 km. The island is located at the north-western edge of the study area (Fig. 1), north of Ranongga Island and south-west of Vella Lavella Island. The microatoll is located at 7.886°S , 156.499°E , measures approximately 3.5 m in diameter and 1.5 m in height (Fig. 9), and sits on a sandy substrate. Neighboring uplifted microatolls at the site show central depressions indicating rapid submergence and subsequent upward and inward growth from their living peripheries over recent decades. The research team collected seven cores (Cores A–H): five vertical and two horizontal. This configuration provides a direct opportunity to test the effect of sampling direction and path dependency on the $\delta^{13}\text{C}$ proxy. Initial observations identify two major die-down events;

shoreline retreat near Supato Village on southern Vella Lavella within one month (Grover, 1965).

6.3 Literature Data Comparison

$\delta^{13}\text{C}$ is a widely used climate proxy with a large body of published datasets that can be reanalyzed from a tectonic perspective. Coral habitats intersect with tectonically active regions in several parts of the world, and examining existing $\delta^{13}\text{C}$ records through a geodetic lens may reveal tectonic signatures that were not the focus of the original studies.

One promising candidate region is the Gulf of Aqaba in the northern Red Sea. Felis et al. (1998) documented 4–5 year rise-and-fall oscillations in coral $\delta^{13}\text{C}$ with amplitudes of approximately $\pm 1\%$. The study site lies in close proximity to the left-lateral Dead Sea Fault, which could plausibly drive the observed long-term $\delta^{13}\text{C}$ variability through periodic vertical displacement rather than climate forcing alone. Additionally, three active GPS stations in the region record vertical motion, providing a critical geodetic reference for any isotope–deformation comparison. Identifying analogous signals in independent published records would strengthen the case for $\delta^{13}\text{C}$ as a broadly applicable paleogeodetic proxy beyond the Solomon Islands setting.

7 Research Package 3: Neotectonic Observations: NISAR

Recent geodetic observations are essential for any neotectonic investigation. The absence of permanent GPS stations on the New Georgia Group (NGG) increases reliance on satellite-based datasets, particularly InSAR. The third research package will therefore focus on the analysis of NISAR data to characterize neotectonic motions across the NGG, with emphasis on any long-term signals that mirror the decadal displacement patterns observed in the $\delta^{13}\text{C}$ record.

Kuo et al. (2016) conducted the first GPS measurement campaign on the NGG between 2011 and 2014, deploying 10 stations across the island group. The SIBO station, located on Simbo Island on the subducting plate, lies only 13 km from stations on the overriding plate, providing a rare opportunity to directly detect interplate coupling from geodetic data spanning both sides of the trench. The observations indicate clear coupling between the upper and lower plates during 2011–2014, with the degree of coupling decreasing at stations located further from the trench. The TEPA station, closest to the 2010 earthquake epicentre, shows notably different behavior. Vertical motion results are consistent with this pattern: nearly all stations record subsidence rates of up to $23.2 \pm 1.4 \text{ mm yr}^{-1}$, while TEPA exhibits uplift at $25.1 \pm 3.3 \text{ mm yr}^{-1}$.

The NISAR satellite was successfully launched and began collecting data from October 2025 onward, though data availability specifically over the NGG remains to be confirmed. NISAR observations over the region will allow the spatial distribution of line-of-sight (LOS) deformation rates to be mapped, which, while not equivalent to pure vertical or horizontal displacements, provide a cost- and time-efficient complement to GPS campaigns. The region is covered by dense tropical vegetation year-round; Kendall et al. (2026) demonstrated that L-band measurements can nevertheless capture significant vertical motions in this environment. Collected data during 2026–2027 will provide fundamental data for my research.

8 Budget

The annual cost of a graduate student at the Jackson School of Geosciences (JSG) is approximately \$120,000, and all admitted students receive five years of guaranteed funding. When external funding is unavailable, students may supplement their support through teaching assistant (TA) appointments. With the financial support of my advisors, I was required to serve as a TA for only one semester during my first two years, which allowed me to make substantial progress on the first part of the dissertation (Section 5). Once this work is submitted to a journal, it will open the opportunity to apply for additional external funding. Conference attendance at AGU and EGU has been and will continue to be supported by the JSG Graduate Professional Development Fund (\$5,000), and additional external funding will be actively sought throughout the program.

Table 1: PhD Research Timeline: 2024–2029

Category	Task / Milestone	2024-25		2025-26		2026-27		2027-28		2028-29	
		Fa	Sp	Fa	Sp	Fa	Sp	Fa	Sp	Fa	Sp
Milestones	Required Classes	X	X	X		X					
	Qualifying Exam				X						
	Dissertation Outline				X						
	Final Defense										X
	Thesis Compilation							X	X	X	X
Prof. Dev.	EGU		X		X		X				
	AGU Fall Meeting					X		X		X	
RP1: $\delta^{13}\text{C}$ -RWD	Sampling & Lab	X	X								
	Age Model		X	X	X						
	Time Series Analysis		X	X	X						
	Draft Manuscript 1				X						
RP2: New Corals & Literature $\delta^{13}\text{C}$	Sampling & Lab					X					
	Literature Work					X	X				
	Time Series Analysis					X	X				
	Draft Manuscript 2					X	X				
RP3: NISAR & Geodetic Comparison	Data Cleaning						X	X			
	Deformation Model							X	X		
	Combined Geodetic Map								X		
	Draft Manuscript 3							X	X	X	

9 Work Plan & Timeline

The projected timeline for the five-year program is summarized in Table 1. Research Package 1 occupies the first two years, covering sampling, laboratory analysis, age modeling, and time-series analysis, with a target manuscript submission in Spring/Summer 2026. Research Package 2 follows in 2026–27, combining new coral sampling with the literature comparison study and a second manuscript. Research Package 3 runs from late 2026 through 2028, with NISAR data cleaning, deformation modeling, and a third manuscript planned for 2027–28. Thesis compilation begins in Fall 2027 and the final defense is scheduled for Spring 2029. Required coursework was concentrated in the first two years, and conference attendances at EGU and AGU are planned throughout the active research phases.

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